

陈子均

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1. The intensity of the sunlight falling on the earth is, for normal incidence and before its passage through the atmosphere, about 1.5 kW m^{-2} . Calculate the electric-field strength.

$$I = \frac{1}{2} c \epsilon_0 E^2 = 1.5 \times 10^3 \text{ W m}^{-2}$$

$$E = \sqrt{\frac{2I}{c \epsilon_0}} = \sqrt{1.13 \times 10^6} = 1063 \text{ V/m}$$

Electric field strength

2. Pulsed high power lasers can be easily focussed to a power density I of 10 GW/cm^2 . Calculate the E and B fields. Compare them with the electric field in an H atom at a distance of one Bohr radius, and the magnetic field on the surface of the earth, respectively.
3. Calculate the number of photon modes in the visible part of the spectrum

$$I = 10 \times 10^9 \text{ W/cm}^2 = 10^{10} \text{ W/m}^2$$

$$I = \frac{1}{2} c \epsilon_0 E^2 \Rightarrow E_1 = 2.74 \times 10^8 \text{ V/m}$$

$$E_H = 5.15 \times 10^{11} \text{ V/m} > E_1$$

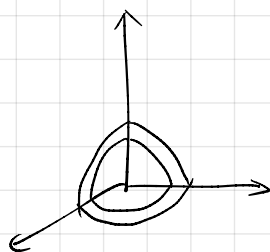
$$B_1 = \frac{E_1}{c} = 0.916 \text{ T}$$

$$E_e = 5 \times 10^{-5} \text{ T} < B_1$$

3. Calculate the number of photon modes in the visible part of the spectrum ($\approx 400 \text{ nm} \lesssim \lambda_v \lesssim 800 \text{ nm}$) in a box of 1 cm^3 .

$$k_x = k_y = k_z = \frac{2\pi}{\lambda} n$$

$$\text{每个模式的体积: } \Delta k_x \Delta k_y \Delta k_z = \left(\frac{2\pi}{\lambda}\right)^3 = \frac{8\pi^3}{\lambda^3} = \frac{8\pi^3}{V}$$



在 $k + dk$ 范围内, 体积为 $4\pi k^2 dk$

$$\Rightarrow dn = 2 \times \frac{4\pi k^2 dk}{\frac{8\pi^3}{V}} = \frac{V k^2}{\pi^2} dk$$

$$400 \text{ nm} \leq \lambda \leq 800 \text{ nm}$$

$$7.85 \times 10^{-3} \text{ nm}^{-1} \leq k = \frac{2\pi}{\lambda} \leq 0.0157 \text{ nm}^{-1}$$

$$N = \int dn = \int_{7.85 \times 10^{-3}}^{0.0157} \frac{V k^2}{\pi^2} dk = 1.145 \times 10^4$$

4. Derive the density of states for electrons in a 3-dimensional semiconductor

$$E_k = E_c + \frac{\hbar^2 k^2}{2m^*}$$

$$\Delta k_x \cdot \Delta k_y \cdot \Delta k_z = \frac{(2\pi)^3}{V}$$

$$N(k) = 2 \times \frac{\frac{4}{3}\pi k^3}{\frac{(2\pi)^3}{V}} = \frac{V}{3\pi^2} k^3$$

$$g(k) = \frac{dN}{dk} = \frac{V}{\pi^2} k^2$$

$$k = \sqrt{\frac{2m^* E}{\hbar^2}} \quad \frac{dk}{dE} = \sqrt{\frac{2m^*}{\hbar^2}} \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{E}}$$

$$g(E) = \frac{dN}{dk} \frac{dk}{dE} = \frac{V}{\pi^2} k^2 \cdot \sqrt{\frac{2m^*}{\hbar^2}} \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{E}}$$

$$= \frac{V}{\pi^2} \cdot \frac{2m^* \cdot E}{\hbar^2} \cdot \sqrt{\frac{2m^*}{\hbar^2}} \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{E}}$$

$$= \frac{V(2m^*)^{\frac{3}{2}}}{2\pi^2 \hbar^3} \sqrt{E}$$

In Unit Volume : $g(E) = \frac{(2m^*)^{\frac{3}{2}}}{2\pi^2 \hbar^3} \sqrt{E}$

$$= \frac{4\pi (2m^*)^{\frac{3}{2}}}{h^3} \sqrt{E}$$